

Disaster Rangers – AI for Crisis Response

Artificial Intelligence Project | 2026

: Deema M. AL-Maqadma,

Supervisors: Dr. Shadia Baroud, Eng. Sara Alzamly

Introduction

-  **General AI Definition:** Leveraging computational systems to mimic human intelligence for complex problem solving.
-  **Course:** Artificial Intelligence Core Project.
-  **Domain:** AI application in high-stakes disaster management scenarios.
-  **Mission Criticality:** Why rapid crisis response is vital for saving lives and infrastructure.

Domain Context



Information Pulse

Social media has become the primary source of real-time data during active disaster events.



The Data Challenge

Significant noise: jokes, metaphors, and irrelevant content hinder actionable intelligence.



Real-World Impact

Early warning systems and resource deployment driven by faster decision-making models.

Project Idea

Detection & Alerting

Building an AI-powered system to distinguish real disaster reporting from casual conversation on Twitter.

Providing emergency teams with reliable, filtered alerts for immediate action.

Multi-Class Classification

Automatically classifying incidents into specific disaster types:

 Fire,  Flood,  Earthquake.

Dataset Overview

Component	Description
Source	Kaggle Disaster Tweets dataset (7,613 entries)
Labeling	Balanced: Disaster (43%) vs No Disaster (57%)
Features	ID, Keyword, Location, Text content, Target label
Quality	Missing data in keyword/location; Text is 100% reliable

NLP & Preprocessing

-  **Cleaning:** Lowercasing, URL/punctuation removal, and stemming.
-  **Stopwords:** Removing non-semantic words to focus on critical intent.
-  **Feature Engineering:** Text length, exclamation count, and capital ratio analysis.

Why NLP is Essential:

To distinguish between metaphorical "on fire" (casual) and physical "on fire" (emergency).

Models & Experiments

Binary Classification

Detecting the presence of a disaster (Yes/No).

- Logistic Regression / SVM
- LSTM / CNN
- FastText

Multi-Class Task

Granular classification of crisis types.

- Earthquake, Flood, Fire
- No Disaster, Other
- Automated Pipeline Evaluation

Results & Insights



📌 *FastText was chosen for real-time deployment due to its optimal balance of speed and accuracy.*

Automation



Rapid Prep

Automated cleaning and feature engineering pipeline.



Auto-Eval

Metrics, confusion matrices, and reports generated instantly.



Scalability

Reproducible workflow designed for high-volume data streams.

Future Work

- 🔗 **Multi-Source Expansion:** Beyond Twitter to Facebook, Instagram, and more.
- 👥 **Multi-Agent Systems:** Collaborative AI agents for regional coverage.
- 🔗 **System API:** Direct integration with emergency dispatch centers.
- 🤖 **Response Chatbot:** AI assistants for users in affected disaster zones.

Conclusion

“*We are prepared to take Disaster Rangers further — expanding to multi-source, multi-agent systems — to save lives through faster AI-powered crisis response.*”

✓ Robust Results | Ready for Deployment